



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (15/06/2026 – 20/06/2026)

WEEK 5

PROJECT NAME

ROAD CONDITION DETECTION

PROJECT DESCRIPTION :

**AUTOMATIC DETECTION AND
SEGMENTATION OF ROAD DAMAGES**

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Date	Task
15/06/2026	Integrated the trained YOLO11 road segmentation and damage segmentation models into a dashboard.
16/06/2026	Updated the inference pipeline with instance segmentation, tracking IDs, and road-region validation logic.
17/06/2026	Improved the dashboard UI, enhanced visualization, and customized class labels, colors, and tracking display.
18/06/2026	Debugged model loading, corrected class mapping, fixed configuration issues, and resolved inference errors.
19/06/2026	Worked on report generation, PDF export, session history, and analytics while fixing Unicode and export-related issues.
20/06/2026	Tested the complete application with videos, verified segmentation results, optimized processing, and fixed remaining dashboard issues.
21/06/2026	Holiday (Sunday)

Next Week Plan	Finalize the RoadSense dashboard by resolving remaining UI and model integration issues, optimize segmentation accuracy and performance, complete report generation features, and prepare the application for demonstration and deployment.
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REFERENCE:

- [1] G. Jocher and J. Qiu, *Ultralytics YOLO11 Documentation*, Ultralytics, 2024. Available: <https://docs.ultralytics.com/models/yolo11>. Accessed: Jun. 2026.
- [2] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.
- [3] A. Petrov, "Streamlit: The Fastest Way to Build Data Apps," Streamlit Documentation. Available: <https://streamlit.io>.

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